Why BLDC ?

Reduced noise, wear, and friction caused by brushes and Torque-Speed characteristics is linear.[Direct-drive robots: theory and practice]

Design

Basic Terminology

Stator & Rotor

Stator, the stationary part, which carries the winding.Rotor, the rotating part, which carries the magnet.

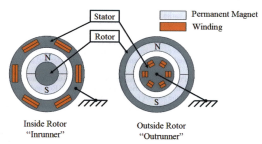

Figure 1: Stator & Rotor, Source:[1]

Air Gap

Air gap is the distance between stator and rotor.

Tooth & Slot

Winding are wounded around the laminated steel structure called tooth which channels more magnetic flux through them. Slot is the section between two tooth. Three phase motor have slots (and teeth) that is evenly divisible by three.

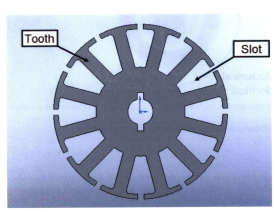

Figure 2: Tooth & Slot, Source:[1]

Phase

A phase is an individual group of windings with a single terminal accessible from outside the motor. Most brushless motors are three-phase.

Turns

Pole (N_m)

A pole is a single permanent magnet pole, north or south. The minimum number of poles is two, but motors can have any even number of poles.The number of poles is not directly related to the number of slots, although there are common combinations of slot and pole counts that work well [6].

Electrical Angle (θ_e) **vs Mechanical Angle** (θ_m)

Mechanical degrees in a motor refers to the rotation of the shaft. 1 revolution of the shaft equals 360 deg mechanical degrees.

When an electrical machine is operating as a motor, the idea is to create a traveling, rotating magnetic field, via the stator so that this moving flux attracts the rotor. Electrical degree is the angle through which magnetic field has rotated, i.e., 360 electrical degree equals transistion form "North" to "South" to "North"

A 2 pole motor has 1 "North" pole and 1 "South" pole on the rotor. So in order for it to turn 360 electrical degrees ("North" to "South" to "North"), it needs to rotate 360 (360) mechanical degrees. A 4 pole motor has 2 "North" poles and 2 "South" poles. That means that 360 electrical degrees will occur when the shaft has rotated only 180(360/2) mechanical degrees. Thus for N_m Poles we have,

$$\theta_e = \frac{N_m}{2}\theta_m$$

Motor Parameters

Torque/Back EMF constant k_t

$k_t = f(\theta_e)$, for a BLDC Motor and for DC Motor $ k_{t} = c $

Motor Constant

Electric Time Constant [Direct-drive robots: theory and practice]

Motor Constant K_m is the ability to produce Toqrue for a given heating power loss. It is winding invariant (as long as same conductimg wires used) but why? (Leanr about turns and wires thickness) (a)[https://ipfs.io/ipfs/QmXoypizjW3WknFiJnKLwHCnL72vedxjQkDDP1mXWo6uco/wiki/Motor_constants.h

$$K_m = \frac{\tau}{\sqrt{P_{loss}}} = \frac{IK_T}{\sqrt{I^2R}} = \frac{K_T}{\sqrt{R}}$$

$$K_m^2 \propto r_{gap}^3$$

[Actuator design for high force proprioceptive control in fast legged locomotion]

Thermal Specific Torque Density K_{ts}

Form, [Design Principles for a Family of Direct-Drive Legged Robots], we have

$$K_{ts} := \frac{K_t}{m}\sqrt{\frac{1}{R_{th}R}}$$

Resistance

Inductance

It will resist the rapid change in current by producing a voltage across it but its not the Back EMF. Back EMF is the only component that is generate by the permanent magnet flux

inductance, to accommodate for this. The value of the synchronous inductance L_s is:

L_a is ?

The winding inductance has many theoretical and practical effects on the motor. It stores energy in the form of a magnetic field any time there is current in the winding. When a winding is switched off, this energy must go somewhere. For this reason, controllers contain "flyback diodes" that allow this current to circulate even when all the switches are open. Under highfrequency pulse-width modulated (PWM) control, the winding inductance also filters out current ripple. However, as a low-pass filter on current it also creates phase lag which is the motivation for the use of field-oriented control

The winding inductance is a function of motor geometry and the number of turns in the winding.

Winding Configuration

They have three phase windings and can be connected to each other in wye or delta configuration. Wye has higher torque (theoretically torque constant is greater by a factor $\sqrt{3}$ [2, p. 25]) because in the wye configuration, at any time one of phase is open and other two-phase are in series thus equal current flows through them whereas in delta it gets divided into two phases. Thus more current in each phase causes more torque, one will push and another phase will pull in the wye configuration.

Saliency

Saliency means the inductance varies with rotor position. [Design of Brushless Permanent-Magnet Machines, p. 68]

Non-Salient

Rotor is rotationally symmetric. No tendency to align with stator if magnets in rotor are removed. i.e., No **Reluctance Torque**, and winding inductance does not vary with rotor position.

Non-salient poles exhibit both attractive & repulsive gap forces -Why ? (lec1)

Field windings & Armature winding ?

Proximity effect

what is reluctance torque?

Electrical Loading [Design of Brushless Permanent-Magnet Machines, p. 68]
Magnetic Loading [Design of Brushless Permanent-Magnet Machines, p. 68]

outrunners (rotor on the outside) will be preferrable to inrunners (rotor on the inside), and this measure is tied favorably to a motor's radius to depth ratio [Direct-drive robots: theory and practice] as well as a large gap radius [Actuator design for high force proprioceptive control in fast legged locomotion]. The measure is fundamentally winding invariant [Leg Design for Energy Management in an Electromechanical Robot], but in practice other details of the motor's construction (especially relating to the stator core and volume of copper) are critical.

[**Direct-drive robots: theory and practice**] Direct Drive has no damping. *Mechanical Time Constant* $\tau_{mech} = \frac{inertia\ load}{mechanical\ damping\ coefficient}$ is large comapred to geared drive.

[Actuator design for high force proprioceptive control in fast legged locomotion]

The gap radius is the distance from the rotating axis to the center of the gap between permanent magnets and the stator.

The area inside of the gap increase proportionally.

$$M \propto r_{gap}$$

$$\tau \propto r_{gap}^2$$

$$J \propto r_{gap}^3$$

$$\frac{\tau}{M} \propto r_{gap}$$

$$\frac{\tau}{J} \propto \frac{1}{r_{gap}}$$

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[Actuator design for high force proprioceptive control in fast legged locomotion]

winding Invarainat? [Leg Design for Energy Management in an Electromechanical Robot]

Winding volume, $V = \pi\rho^2 L$.

The resistance of the winding,

$$R \propto \frac{L}{A} \propto \frac{1}{\rho^4}$$

$$K_t \propto L \propto \frac{1}{\rho^2}$$

$$\tau = K_t i \implies \tau \propto \frac{i}{\rho^2}$$

$$P_{loss} = i^2 R \propto \frac{i^2}{\rho^4}$$

thus

$$K_m = \frac{T}{\sqrt{P_{loss}}} \propto \rho^0$$

[Compliance and Force Control for Computer Controlled Manipulators] Force Control and Position Control are two ends of extreme. Force control sensible in case of contact. Position control sensisble in case of contact-less

Thus for a real world we need a hybrid approach

Losses

Inefficiencies can be grouped into two general categories:

1. Torque/Copper/I^2 Losses
2. Speed/Iron/E^2 Losses

Torque/Copper/I^2 Losses: Winding Resistance, Power Ratings

They are due to resistance heating of the windings, and are proportional to the square of the output torque.

Speed/Iron/E^2 Losses: Delta Windings, Eddy Current, and Mechanical Losses

Iron Core Losses = Eddy Curremt Losses + Hystersis Losses

Hysteresis in the magnetic domains in the core appears as a constant friction torque, while the eddy current torque is proportional to speed.

Selection Criteria

[Actuator design for a high performance legged biped robot]

A smaller motor will have lower iron losses at a given speed (less iron to create drag), and a larger motor will have lower copper losses for a given torque.

One interesting approach to motor sizing for a robot is to choose a small motor sized for walking, and then to overload it heavily during more strenuous activities.

This might result in huge copper losses due to squaring the high winding current, but for brief periods the effect on overall efficiency is minimal.

The thermal mass of the motor windings is high enough that it might not burn up for several seconds, and if it is cooled well enough the power level may be sustainable for longer activities.

This is particularly true for permanent magnet brushless motors, in which the stator coils are stationary and in contact with the case, well-positioned for heat sinks or water cooling jackets. (For brushed motors the magnets are on the outside and the windings are in the rotor, leaving few options for cooling.

Drive

References

[1] S. W. Colton, "Design and prototyping methods for brushless motors and motor control," Masters Thesis, Massachusetts Institute of Technology, 2010 [Online]. Available: http://dspace.mit.edu/handle/1721.1/61599

[2] S. Kalouche, "Design for 3d agility and virtual compliance using proprioceptive force control in dynamic legged robots," Masters Thesis, Carnegie Mellon University, 2016.